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Original Research Article

Perioperative transcutaneous electrical acupoint stimulation for improving postoperative gastrointestinal function: A randomized controlled trial



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ABSTRACT

Background: Postoperative gastrointestinal dysfunction (PGD) is one of the most common complications in patients undergoing major abdominal surgery. Acupuncture has been used widely in gastrointestinal diseases due to its effectiveness and minimally invasive nature.

Objective: This study evaluated the efficacy of using transcutaneous electrical acupoint stimulation (TEAS) during the surgery and postoperative recovery in patients with gastric and colorectal surgery for improving postoperative gastrointestinal function.

Design, setting, participants and interventions: A total of 280 patients undergoing abdominal surgery were stratified by type of surgery (i.e., gastric or colorectal surgery) and randomly allocated into the TEAS group (group T) or the sham group (group S). Patients in group T received TEAS at LI4, PC6, ST36 and ST37. Patients in group S received pseudo-TEAS at sham acupoints. The stimulation was given from 30 min before anesthesia until the end of surgery. The same treatment was performed at 9 am on the 1st, 2nd and 3rd days after surgery, until the recovery of flatus in patients.

Main outcome measures: The primary outcome was the time to the first bowel motion, as detected by auscultation. The secondary outcomes included the first flatus and ambulation time, changes of perioperative substance P (SP), incidence of PGD, postoperative pain, postoperative nausea and vomiting (PONV) and some economic indicators.

Results: The time to first bowel motion, first flatus and first ambulation in group T was much shorter than that in group S ($P < 0.01$). In patients undergoing colorectal surgery, the concentration of SP was lower in group T than in group S on the third day after the operation ($P < 0.05$). The average incidence of PGD in all patients was 25%, and the frequency of PGD was significantly lower in group T than in group S (18.6% vs. 31.4%, respectively; $P < 0.05$). TEAS treatment (odds ratio = 0.498; 95% confidence interval: 0.232–0.786) and type of surgery were relevant factors for the development of PGD. Postoperative pain score and PONV occurrence were significantly lower in group T ($P < 0.01$). Postoperative hospitalization days and the resulting cost to patients were greatly reduced in the TEAS group ($P < 0.01$).

Conclusion: Perioperative TEAS was able to promote the recovery of postoperative gastrointestinal function, reduce the incidence of PGD and PONV. The concentration of SP was decreased by TEAS treatment, which indicates that the brain-gut axis may play a role in how TEAS regulates gastrointestinal function.

Trial registration: Chinese Clinical Trial Registry, ChiCTR1900023263.

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1. Introduction

Postoperative gastrointestinal dysfunction (PGD) is one of the most common complications among patients who have undergone major abdominal surgery [1]. Despite attempts to minimize its prevalence, the incidence of PGD is still high, affecting between 10% and 30% of patients who have received abdominal surgery [2,3]. The delayed recovery of gastrointestinal function is associated with a longer hospital stay and higher risks of postoperative complications, which not only increases the healthcare costs to patients but also increases the burden on hospital resources [4,5]. These costs emphasize the importance of prevention and treatment of PGD.

Recently, various techniques have been investigated to prevent and alleviate PGD. Early movement [6] and reducing opiate use [7] in favor of non-steroidal anti-inflammatory drugs (NSAIDs) [8] are actions that can facilitate the recovery of gastrointestinal function. However, patients are often too weak to move early, and the side effects of NSAIDs can limit their application. Thus, there is an urgent need to explore an efficient and practicable method to manage PGD. Acupuncture is a promising option that may be able to meet these needs.

As an important part of traditional Chinese medicine, acupuncture is widely used in clinical practice to treat a variety of diseases and physiological disorders [9]. Transcutaneous electrical acupoint stimulation (TEAS) is a contemporary modification of traditional acupuncture that sends electrical impulses into acupoints through electrodes on the skin surface; this method has gained increasing acceptance in China and abroad [10,11]. A recent cross-sectional study showed that gastrointestinal disease is one of the most common complaints treated with acupuncture in the United States [12]. However, the effect of providing TEAS during surgery and as a part of postoperative care on the recovery of gastrointestinal function, as well as its underlying mechanism of action, is still unknown.

In recent years, a growing number of investigations have put a spotlight on the brain-gut axis, which is believed to connect the central nervous system (CNS) and the gastrointestinal tract [13,14]. The brain affects the sensation, motility and endocrine release of the gastrointestinal tract through signals, and visceral messages also affect brain function [13]. Brain-gut peptides, distributed along the gastrointestinal tract and in the CNS, are involved in regulating the gastrointestinal function [15]. Strikingly, recent investigations have found that acupuncture at different acupoints could specifically cause imaging changes in different brain regions [16]. Hence, this study observed changes of perioperative substance P (SP), a type of brain-gut peptide, in patients who underwent gastrointestinal surgery.

This study investigates the effects of providing TEAS during surgery and as a part of postoperative care on the recovery of gastrointestinal function in patients who have undergone gastrointestinal surgery, and it evaluates whether the underlying mechanism of this action is related to the brain-gut axis.

2. Methods

2.1. Ethics statement

This single-blinded randomized controlled trial was conducted at Beijing Friendship Hospital, Capital Medical University, China. The study protocol was approved by Beijing Friendship Hospital Ethics Committee (No. 2019-P-065-01). The study was registered in the Chinese Clinical Trial Registry (No. ChiCTR1900023263). The protocol of this study was published on *Trials* in 2020 [17].

2.2. Inclusion and exclusion criteria

Patients who met the following criteria were considered to be eligible for the trial: aged 18–70 years with no gender restriction; American Society of Anesthesiologists Physical Status Grades I–III [17,18]; body mass index (BMI) between 18 kg/m² and 30 kg/m²; undergoing endotracheal intubation and laparoscopic abdominal tumor resection under general anesthesia (stomach, colon or rectum); using postoperative analgesia pump; volunteering to participate in this study and sign the informed consent.

Patients meeting the following criteria were excluded from the study: surgical incision or scar near the acupoints of Hegu (LI4), Neiguan (PC6), Zusanli (ST36) and Shangjuxu (ST37); local skin infection at study acupoints; nerve damage in upper or lower limbs; a history of spinal surgery; participation in other clinical trials within the last four weeks; inability to understand the Visual Analog Scale (VAS) score or disagreeing with the use of patient-controlled analgesia; presence of a pacemaker; patients with severe CNS diseases or severe mental disorders; operations requiring enterostomy or converts to laparotomy; patients who need to be transferred to the intensive care unit (ICU) for treatment after surgery.

Drop-out criteria were as follows: violation of the test program; occurrence of serious adverse events or the subjects requested to withdraw during the test.

2.3. Randomization and blinding

After confirming eligibility and providing informed consent, patients were stratified by type of surgery (i.e., gastric or colorectal surgery) and randomly assigned to the TEAS group (group T) or the sham group (group S). This procedure was conducted using the blocked randomization approach with a block length of four. Then, computer-generated random numbers were used to determine the allocation of blocks. The randomization schedule was kept in opaque sealed envelopes, which were opened by an independent investigator who was not an assessor in this study. The technicians who carried out acupuncture therapy and anesthesia knew the grouping information. While observers responsible for postoperative follow-up and participants were blinded to the grouping information. The trial adhered to establish procedures to maintain separation between the outcome assessors and the technicians. The technicians did not participate in outcome measurements.

2.4. Interventions

This is a single-center study and all patients recruited were treated at Beijing Friendship Hospital, which is affiliated to Capital Medical University. All participants underwent general preoperative visits after randomization. Once in the operating room, standard monitoring, including electrocardiogram, oxygen pulse saturation, non-invasive blood pressure and end-tidal carbon dioxide partial pressure, was established for all patients. Acupuncture treatments were performed by an acupuncturist from the Department of Traditional Chinese Medicine.

In addition to routine anesthesia, patients in group T who received acupuncture therapy were treated bilaterally at two groups of distal acupoints: one group consisted of Hegu (LI4) and Neiguan (PC6), the other contained Zusanli (ST36) and Shangjuxu (ST37; Fig. 1). The skin of the bilateral acupoints was disinfected with 75% alcohol, and after drying, self-adhesive electrodes were attached to the site of these acupoints, rather than inserted into the skin. They were connected to two guide wires, which were attached to a HANS acupoint nerve stimulator (HANS-200A, Nanjing Jisheng Medical Technology Co., Ltd., China; Fig. 2); each patient required 8 electrodes. Stimulation was delivered with a

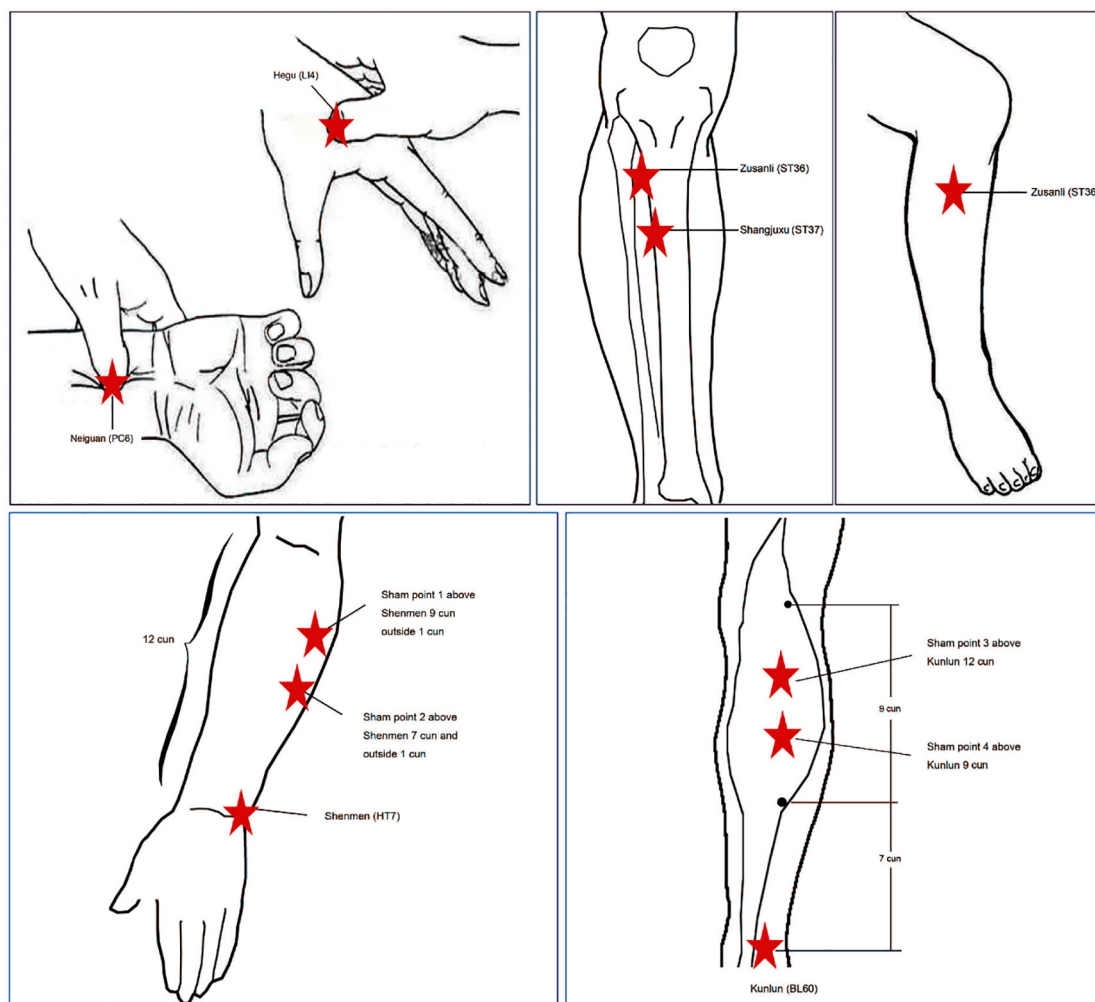


Fig. 1. The location of acupoints. Cun is a unit of length.

disperse-dense wave, 2 and 100 Hz in frequency, alternating every 3 s. The wave widths were 2 Hz for 0.6 ms and 100 Hz for 0.2 ms, and the waveform was complete symmetric biphasic pulse. The stimulation intensity was adjusted in accordance with the maximal level tolerated by each patient.

Group S was a sham acupuncture group, in which patients were treated bilaterally at four sham acupoints of LI4, PC6, ST36 and ST37. These four pseudo-acupoints were located 1 cun ulnar-lateral, and 7 cun and 9 cun proximal to Shenmen (HT7) (Fig. 1); and 9 cun and 12 cun proximal of Kunlun (BL60) (Fig. 1). These sham points are picked because there are no meridians or channels through these four sites. After sterilizing as above, electrodes were attached to these sham acupoints and connected to two guide wires that were attached to a HANS acupoint nerve stimulator. However, no stimulation was delivered.

According to the results of our previous study [18] and other studies [11], TEAS application began before operation. The stimulation (real or sham) lasted from 30 min before surgery until the end of the operation. At 9:00 am on the 1st, 2nd and 3rd days after surgery, the same acupuncturist that attended the patients during their operation performed the same acupuncture manipulation (real or sham) that they had received during surgery. Each session lasted for 30 min until the flatus of patient recovered. Patients in groups T and S received the same postoperative management.

All patients received same anesthesia and postoperative treatment; the only difference in treatment among groups was the loca-

tion of real or sham acupoints. Patients received general anesthesia intravenously, as determined by the chief anesthesiologist. Postoperative analgesia measures included an intravenous analgesic pump that dispensed 250 μ g sufentanil in 250 mL saline. If the VAS score was ≥ 4 points, 50 mg flubiprofen (intravenously guttae) was given. The occurrence of nausea and vomiting was also recorded.

2.5. Outcome measures

All the preoperative general status and surgical variables, including age, gender, height, weight, comorbidities, the history of abdominal surgery, intraoperative infusion volume, the duration of the anesthesia and the operation, were monitored and recorded. Postoperative data were collected by a blinded observer.

The primary outcome was the time to the first bowel motion by auscultation. The bowel sounds were listened for by a surgical resident every 4 h, after the operation. Time to first bowel motion was defined as the time from the end of the operation to the commencement of regular bowel sounds (more than two sounds every minute) [19].

The secondary outcomes included the time to first flatus and first ambulation, the level of perioperative plasma SP, the occurrence rate of PGD, the postoperative pain scores, the incidence of postoperative nausea and vomiting (PONV) and some economic indicators. The time to first flatus and first ambulation was

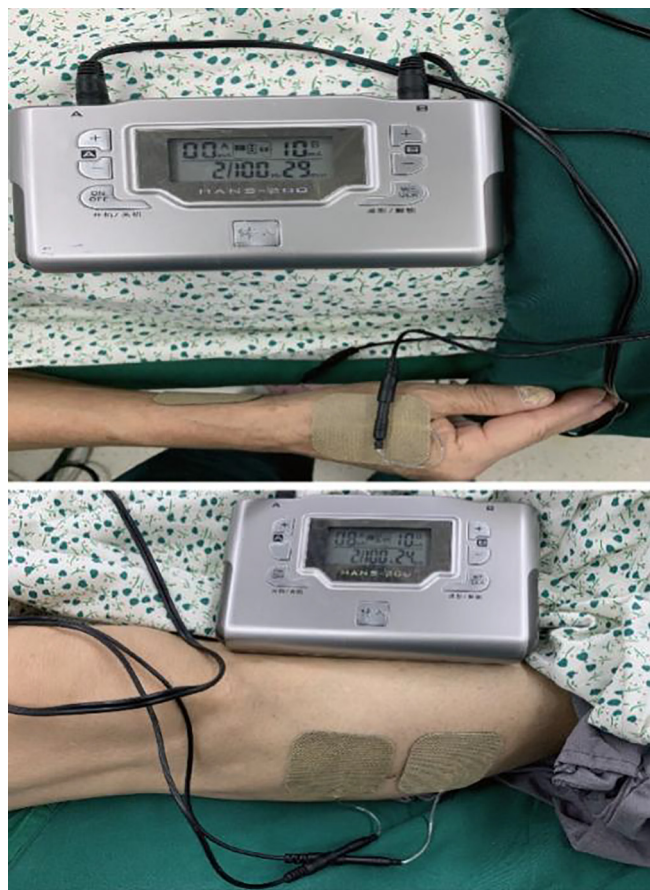


Fig. 2. The connection between acupoints and the HANS acupoint nerve stimulator.

obtained from the patients or their caregivers. On the morning of the day of surgery, and the first and third day after surgery, venous blood samples were collected from patients to detect SP level. SP kits were obtained from RayBiotech (Guangzhou, China), and their use strictly followed the manufacturer's instructions. Absence of bowel sounds for more than 48 h after surgery in this study is considered to suggest the occurrence of PGD [7,20]. The extent of postoperative pain was assessed using the VAS, with 0 indicating no pain and 10 indicating the worst pain imaginable. The VAS value and PONV were recorded at 6 and 12 h after surgery, as well as at 9:00 am and 3:00 pm on postoperative day 1 and day 2. Economic indicators were length of hospital stay and the cost of hospitalization. The total number of days from the date of surgery to discharge was recorded as length of hospital stay.

2.6. Sample size calculation

We evaluated the effectiveness of TEAS according to our pre-experimental results. We used the COX regression sample size calculation formula. We set unilateral $\alpha = 0.05$, power = 0.90 and hazard ratio = 1.5, using PASS 2011 software (NCSS, USA) to estimate the necessary sample size, and calculated that a total of 256 effective cases were needed. Considering a 10% loss to follow-up, the sample size was eventually set at 280 cases, including 140 cases in the TEAS group and 140 cases in the sham TEAS group.

2.7. Statistical analysis

SPSS statistical software (version 25.0, IBM, USA) was used for data analysis and to calculate descriptive statistics. Normally dis-

tributed quantitative data were represented as mean $\pm$ standard deviation, and *t*-tests were used to assess the differences between the treatment group and the control group. Non-normally distributed quantitative data were represented with their median and inter-quartile range, and the statistical analysis was performed using the Mann-Whitney *U* test. Qualitative data were represented as numbers and percentages, and intergroup differences were analyzed using the chi-square or Fisher's exact probability test. A *P* value < 0.05 was considered to represent statistically significant differences. The SP value was analyzed by repeated measure analysis of variance. Univariate and multivariate logistic regression analyses were used to analyze the risk factors for postoperative gastrointestinal dysfunction. The database type was per protocol set. The independent variables included in this study were based on previous investigations [21,22].

3. Results

3.1. Baseline characteristics

From September 1st, 2019 to July 20th, 2020, a total of 311 patients were assessed for eligibility; of these, 5 patients were excluded because they did not sign the informed consent. As a result, 306 patients were stratified by type of surgery, gastric surgery ($n = 105$) and colorectal surgery ($n = 201$), and were randomly assigned to group T or group S. Of patients undergoing gastric surgery, one patient in group T and one patient in group S were transferred to the ICU for further treatment after surgery. Further, the operation was cancelled for two patients in the group T and one patient in group S. Among the patients undergoing colorectal surgery, 9 from group T and 7 from group S, were transferred to the ICU for further treatment after surgery. There were 2 cases in group T and 3 in group S, where the operation was cancelled. Thus, a total of 280 patients finally completed the entire trial. An explanatory flow chart is depicted in Fig. 3.

The baseline characteristics of the 280 patients are provided in Table 1, and there was no difference between the subgroups. Patient demographics, intraoperative infusion volume and duration of anesthesia and surgery were similar between the two groups ($P > 0.05$). There were also no statistically significant differences in comorbidities and history of abdominal surgery between the two groups ($P > 0.05$).

3.2. Assessment of gastrointestinal function recovery after surgery

The results showed that postoperative time to first bowel motion in the TEAS group was significantly shorter than that in the S group ($P < 0.01$). In addition, there was a significant difference in the time to the first flatus between the two groups, and the flatus time in group T was much shorter than that in group S ($P < 0.01$). Correspondingly, the time to first ambulation after surgery was shorter in the TEAS group ($P < 0.01$, Table 2).

3.3. Association between factors and PGD

According to our definition of PGD, the time to first bowel sound was more than 2 d, and 70 of the 280 patients (25%) developed PGD. The incidence of PGD in group S was 31.4% (44/140), which was significantly higher than in group T (26/140; 18.6%, $P < 0.05$; Table 3).

All perioperative factors that may affect the occurrence of PGD were analyzed using univariate logistic regression analysis. The results are shown in Fig. 4. In terms of the occurrence of PGD, the differences in intraoperative infusion volume, duration of anesthesia, type of surgery and TEAS treatment were statistically signif-

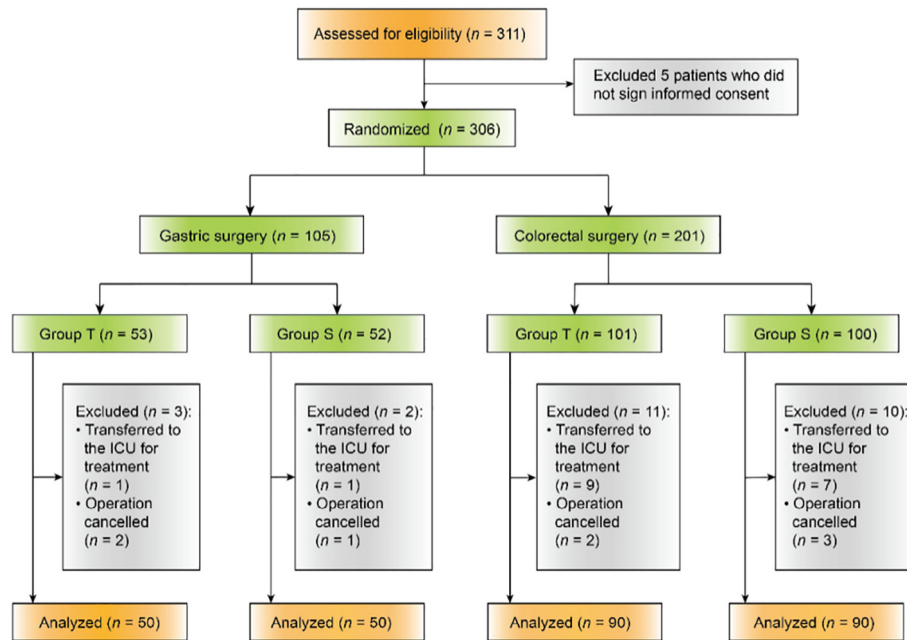


Fig. 3. Flowchart of patient allocation. Group T: transcutaneous electrical acupoint stimulation group; Group S: sham group; ICU: intensive care unit.

Table 1
Baseline characteristics of patients.

Item	All participants			The gastric surgery group			The colorectal surgery group		
	Group T (n = 140)	Group S (n = 140)	P value	Group T (n = 50)	Group S (n = 50)	P value	Group T (n = 90)	Group S (n = 90)	P value
Age (mean ± SD, year)	60 ± 13	62 ± 12	0.168	58 ± 13	59 ± 11	0.630	60 ± 13	63 ± 12	0.175
Sex (male/female, case)	85/55	96/44	0.169	26/24	31/19	0.224	59/31	65/25	0.334
Height (mean ± SD, cm)	166.7 ± 8.0	167.2 ± 7.1	0.539	166.0 ± 7.5	167.6 ± 7.1	0.269	165.9 ± 13.6	167.0 ± 7.2	0.507
Weight (mean ± SD, kg)	67 ± 12	66 ± 10	0.268	67 ± 13	64 ± 10	0.164	67 ± 11	67 ± 10	0.753
Comorbidities (case [%])									
Hypertension	55 (39.3)	48 (34.3)	0.386	24 (48.0)	15 (30.0)	0.065	31 (34.4)	33 (36.7)	0.755
Anemia	13 (9.3)	10 (7.1)	0.514	6 (12.0)	4 (8.0)	0.505	7 (7.8)	6 (6.7)	0.773
Previous abdominal surgery (case [%])	10 (7.1)	6 (4.3)	0.303	3 (6.0)	1 (2.0)	0.617	7 (7.8)	5 (5.6)	0.550
Duration of anesthesia (mean ± SD, min)	246 ± 93	254 ± 83	0.410	270 ± 107	292 ± 92	0.274	232 ± 81	233 ± 70	0.907
Duration of surgery (mean ± SD, min)	186 ± 87	194 ± 79	0.404	216 ± 103	232 ± 88	0.411	169 ± 72	173 ± 65	0.690
Intraoperative infusion volume (mean ± SD, mL)	1940 ± 720	2018 ± 618	0.329	2143 ± 780	2198 ± 555	0.691	1827 ± 650	1918 ± 631	0.339

Group T: transcutaneous electrical acupoint stimulation group; Group S: sham group. SD: standard deviation.

Table 2
Comparison of TBM, TFF and TFA between 2 groups (h, median [IQR]).

Group	TBM	TFF	TFA
Group T (n = 140)	31 (18.8–45.3)**	44 (26.5–63.9)**	44 (24.5–70.7)**
Group S (n = 140)	42 (22.1–52.4)	64 (30.0–70.6)	67 (42.6–72.9)

IQR: interquartile range; TBM: postoperative time to first bowel motion; TFF: postoperative time to first flatus; TFA: postoperative time to first ambulation; Group T: transcutaneous electrical acupoint stimulation group; Group S: sham group. **P < 0.01, vs. group S.

icant ($P < 0.05$). No association was found between the incidence of PGD and age, gender, BMI, hypertension, anemia, previous abdominal surgery, intraoperative dosage of sufentanil, infusion volume and sufentanil dosage 48 h after operation ($P > 0.05$).

Multivariate logistic regression analysis was conducted on factors with $P < 0.1$. Intraoperative infusion volume and duration of anesthesia were modeled as continuous variables in this analysis. The results showed that the type of surgery was a risk factor for PGD. The risk of PGD was significantly higher among patients who had colorectal surgery than those who had gastric surgery (odds ratio [OR] = 28.801; 95% confidence interval [CI]: 6.783–

Table 3
Analysis of the incidence of PGD (case [%]).

Group	Gastric surgery (n = 100)	Colorectal surgery (n = 180)	Sum-up
Group T (n = 140)	0 (0%)	26 (18.6%)	26 (18.6%)*
Group S (n = 140)	2 (1.4%)	42 (30.0%)	44 (31.4%)
Sum-up	2 (2.0%)	68 (37.8%)	70 (25.0%)

PGD: postoperative gastrointestinal dysfunction; Group T: transcutaneous electrical acupoint stimulation group; Group S: sham group. *P < 0.05, vs. group S.

122.295; $P < 0.05$; Table 4). In addition, TEAS treatment was a protective factor associated with reduced incidence of PGD. Patients in the TEAS group had a lower risk of PGD than those in the sham group (OR = 0.427; 95% CI: 0.232–0.786; $P < 0.05$).

3.4. Comparison of plasma concentration of SP

There was no significant difference in the level of preoperative plasma SP, regardless of the type of surgery ($P > 0.05$). Among patients undergoing gastric surgery, the difference in SP levels between group T and group S did not reach statistical significance

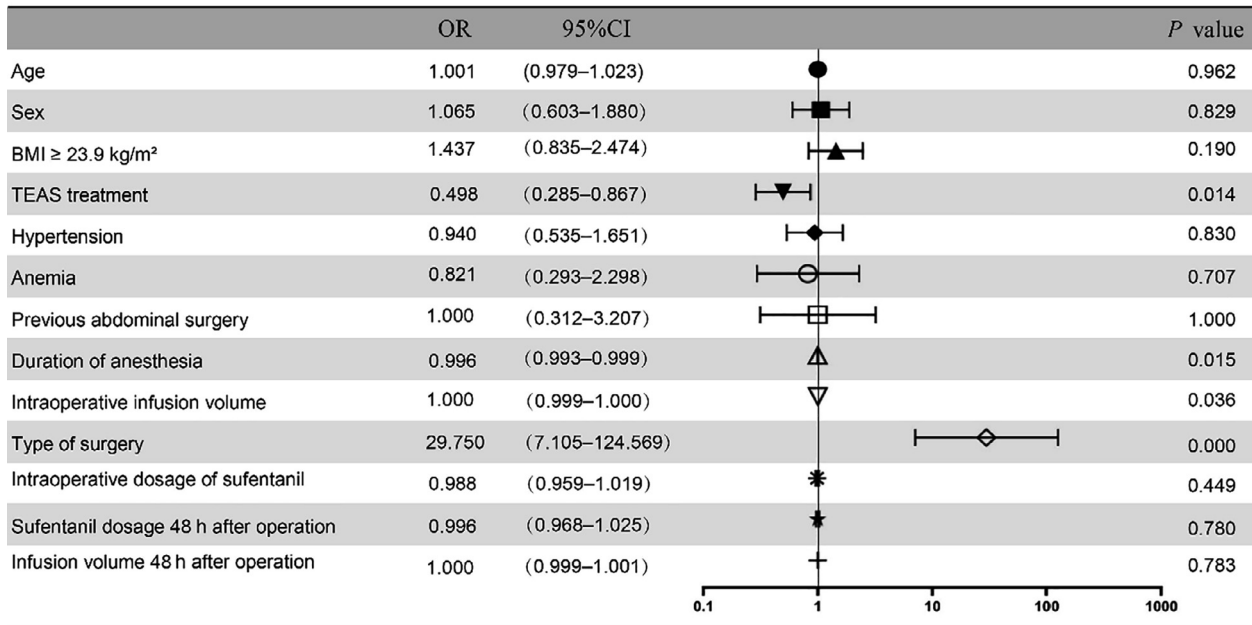


Fig. 4. Forest plot of the results of univariate logistic regression analysis. All perioperative factors were treated as independent variables respectively; the occurrence of PGD was taken as dependent variable for univariate logistic regression analysis. OR: odds ratio; CI: confidence interval; BMI: body mass index; TEAS: transcutaneous electrical acupoint stimulation.

Table 4

Results of multivariate logistic regression analysis of risk factors for PGD.

Item	OR (95% CI)	P value
TEAS treatment	0.427 (0.232–0.786)	0.006
Type of surgery	28.801 (6.783–122.295)	0.000
Duration of anesthesia	0.999 (0.994–1.004)	0.647
Intraoperative infusion volume	1.000 (0.999–1.000)	0.583

PGD: postoperative gastrointestinal dysfunction; OR: odds ratio; CI: confidence interval; TEAS: transcutaneous electrical acupoint stimulation.

at the first and the third days after surgery ($P > 0.05$). In patients undergoing colorectal surgery, the level of SP was lower in group T than in group S at postoperative day 3 ($P < 0.05$, Fig. 5).

3.5. Postoperative pain scores and the incidence of PONV

The VAS score in group T at 6 h after surgery was much lower than in group S (VAS ≥ 4; 21.4% vs. 35.7%; $P < 0.01$). The difference in VAS score between the two groups did not reach statistical significance at the other five postoperative periods ($P > 0.05$). The numbers of cases of PONV that occurred in group T at 6 h postoperative, 12 h postoperative, and 3:00 pm on day 1 were significantly lower than those in group S ($P < 0.01$, Table 5).

3.6. Economic indicators

There were significant differences in economic indicators between the two groups (Table 6). The TEAS group had a significantly shorter postoperative hospitalization time than the sham group ($P < 0.01$). Correspondingly, the TEAS group had a significantly lower average cost of hospitalization than the sham group ($P < 0.01$).

4. Discussion

In this study, our data suggested that for patients undergoing gastrointestinal surgery under general anesthesia, pretreatment

with TEAS improved the postoperative recovery of gastrointestinal motility, decreased the incidence of PGD, relieved postoperative pain and reduced PONV occurrence. More importantly, the effectiveness of TEAS may have been produced through regulating the expression of SP, a brain-gut peptide. In addition, this treatment could correspondingly shorten the length of hospital stay and reduce the cost of hospitalization.

Four acupoints, including LI4, PC6, ST36 and ST37, were selected in this study based on previous research and traditional Chinese medicine Zangfu theory. Many studies [6,23] including our own [18] have found that stimulating LI4, PC6, ST36 and ST37 can improve intestinal motility and regulate endocrine function to reduce gastric acid, which is helpful for adjusting gastrointestinal function. Therefore, we chose these four acupuncture points in this clinical study. Additionally, we used sham controls to explore the role of acupoints stimulation.

PGD is a common postoperative complication of major surgery. As there is no single currently accepted standardized definition of PGD, its incidence is difficult to estimate. Bowel sounds are heard as a rumbling, growling or gurgling noise produced by movement of the contents in the gastrointestinal tract through the small bowel by a series of muscle contractions called peristalsis. The recovery of bowel sounds traditionally suggests the return of postoperative gastrointestinal motility function [20], so in the present study we chose it to evaluate the incidence of PGD. As known, following major abdominal surgery, motility of the stomach and the small bowel is usually restored within 24–48 h [7]. Thus, the absence of bowel sounds 48 h after surgery is considered as a sign of PGD occurrence. In line with previous research [2,3], our data showed a 25% incidence rate. More importantly, we found that pre-treating patients with TEAS during the perioperative period can significantly decrease the incidence of PGD, and the acupuncture intervention was a protective factor associated with PGD, according to logistic regression analysis. This result is expected to provide a new strategy for prevention of PGD.

So far, the specific mechanism through which acupuncture works to relieve gastrointestinal dysfunction is still unclear. In recent years, more and more studies have focused on the brain-

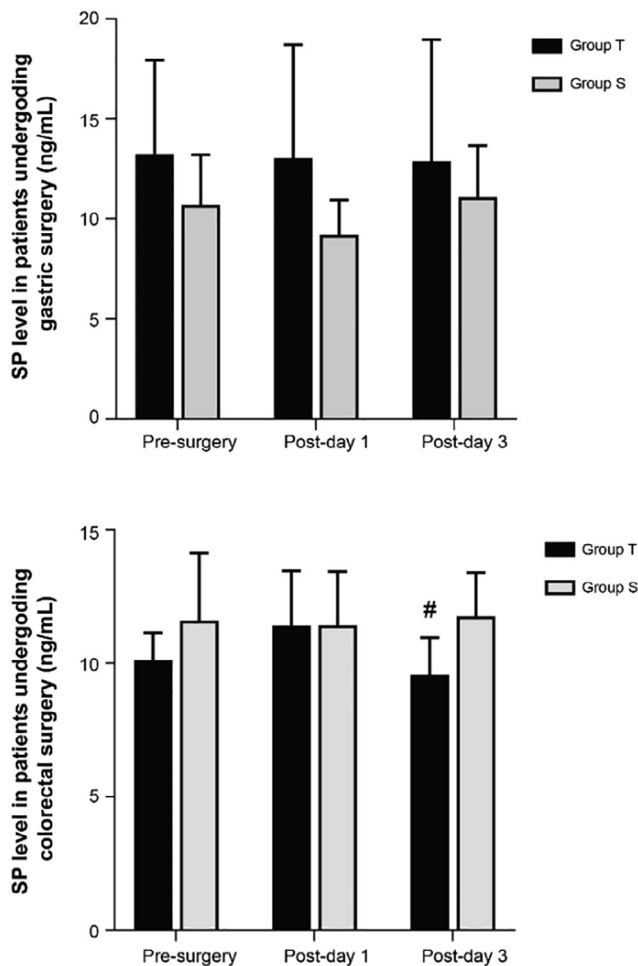


Fig. 5. The level of plasma SP during perioperative period in groups T and S. A: gastric surgery; the data were expressed as mean $\pm$ standard deviation. B: colorectal surgery; the data were expressed as median and interquartile range. #: $P < 0.05$, vs. group S. SP: substance P. Group T: transcutaneous electrical acupoint stimulation group; Group S: sham group.

gut axis [13,14]. Brain-gut peptides are small-molecule peptides that are distributed in both the gastrointestinal tract and CNS, and they are essential for modulating gastrointestinal function [15]. SP, a member of the tachykinin family, is involved in many physiological processes of the gastrointestinal system, and excessive release of SP is reported to cause gastrointestinal dysfunction [24]. In the present study, our data showed that colorectal surgery is more likely to cause PGD than gastric surgery. Moreover, we found that in patients undergoing colorectal surgery, TEAS treatment was able to down-regulate the level of SP and promote post-

Table 6

Postoperative hospitalization days and the cost in two groups (mean $\pm$ standard deviation).

Group	Length of postoperative hospital stay (d)	Hospitalization costs (RMB)
Group T (n = 140)	6.2 $\pm$ 3.3**	64967.5 $\pm$ 27937.2**
Group S (n = 140)	8.1 $\pm$ 3.6	73281.1 $\pm$ 23850.7

Group T: transcutaneous electrical acupoint stimulation group; Group S: sham group. ** $P < 0.01$ vs. group S.

operative recovery of gastrointestinal function. These results support the conclusion that perioperative TEAS treatment can relieve PGD by regulating the brain-gut axis. However, this finding must be investigated further.

In this study, perioperative TEAS treatment decreased the rate of patients reporting VAS scores ≥ 4 at 6 h after surgery this result is similar to our previous work in neurosurgery [18]. The unique feature of this study was the continuous application of TEAS stimulation during and after operation. However, the adjuvant analgesic efficacy of TEAS was not as persistent as we anticipated. This situation may be related to the use of a continuous intravenous analgesic pump in both groups. However, continuous treatment of TEAS after operation was able to significantly reduce the occurrence of PONV. This result was similar to other studies [25,26], which further clarified the effect of continuous application of TEAS in improving gastrointestinal function after gastrointestinal surgery.

This study has some limitations. First, at present, there is no internationally recognized definition for PGD. In this study, we set the definition of PGD as the time to first bowel sound being longer than 2 days. Although this definition has a theoretical basis and support in the literature, it still lacks clear imaging or clinical diagnosis. Therefore, we only use this result as a secondary outcome indicator. Second, we monitored changes in SP levels, but there are many kinds of brain-gut peptides. The effect of TEAS treatment on the other brain-gut peptides should be included in future studies.

5. Conclusions

Results of this study showed that continuous preventive TEAS treatment during the perioperative period was able to promote the recovery of postoperative gastrointestinal function, reduce the incidence of PGD, decrease the occurrence of PONV, relieve postoperative pain, shorten hospital stay and consequently reduce hospitalization costs. The brain-gut axis may be involved in the mechanism through which TEAS operates to regulate gastrointestinal motility. Given that PGD is still a clinically and economically major consequence following gastrointestinal surgery, and that conventional treatments cannot effectively meet the requirements of patients, our study may provide a feasible strategy for preventing the occurrence of PGD.

Table 5

VAS scores and PONV incidence at different postoperative time points in two groups (case [%]).

Time	Postoperative VAS score ≥ 4			PONV occurred		
	Group T	Group S	P value	Group T	Group S	P value
6 h post-OP	30 (21.4)	50 (35.7)	0.008*	51 (36.4)	100 (71.4)	0.000**
12 h post-OP	45 (32.1)	54 (38.6)	0.261	54 (38.6)	81 (57.9)	0.001**
9 am on day 1	14 (10.0)	6 (4.3)	0.063	81 (57.9)	66 (47.1)	0.073
3 pm on day 1	26 (18.6)	25 (17.9)	0.877	36 (25.7)	66 (47.1)	0.000**
9 am on day 2	4 (2.9)	0 (0.0)	0.122	50 (35.7)	56 (40.0)	0.460
3 pm on day 2	7 (5.0)	7 (5.0)	1.000	55 (39.3)	55 (39.3)	1.000

VAS: Visual Analog Scale; PONV: postoperative nausea and vomiting; OP: operation; Group T: transcutaneous electrical acupoint stimulation group; Group S: sham group.

** $P < 0.01$ vs. group S.

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Authors' contribution

ALX designed the hypotheses and the experiments, and LWJ and GC performed the experiments and their analysis. JYW was responsible for data collection. DY was responsible for acupuncture points teaching and TEAS manipulation. XFS was responsible for the English revision. All authors contributed to the scientific discussion of the data and of the manuscript.

Declaration of competing interest

The authors declare that they have no competing interests. please add the content in a separate paragraph.

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References

- [1] Liu L, Yuan X, Yang L, Zhang J, Luo J, Huang G, et al. Effect of acupuncture on hormone level in patients with gastrointestinal dysfunction after general anesthesia: a study protocol for a randomized controlled trial. *Medicine (Baltimore)* 2020;99(14):e19610.
- [2] Chen Y, Dong C, Lian G, Li D, Yin Y, Yu W, et al. Dexamethasone on postoperative gastrointestinal motility: a placebo-controlled, double-blinded, randomized controlled trial. *J Gastroenterol Hepatol* 2020;35(9):1549–54.
- [3] Yuan HC, Xiang Q, Zhang N, Qin WJ, Cai W. Acupuncture combined with early enteral nutrition on patients with postoperative laparoscopic common bile duct exploration: a prospective randomized trial. *Chin J Integr Med* 2019;26(10):769–75.
- [4] Liu Y, May BH, Zhang AL, Guo X, Lu C, Xue CC, et al. Acupuncture and related therapies for treatment of postoperative ileus in colorectal cancer: a systematic review and meta-analysis of randomized controlled trials. *Evid Based Complement Alternat Med* 2018;2018:3178472.
- [5] Murphy MM, Tevis SE, Kennedy GD. Independent risk factors for prolonged postoperative ileus development. *J Surg Res* 2016;201(2):279–85.
- [6] Hou L, Xu L, Shi Y, Gu F. Effect of electric acupoint stimulation on gastrointestinal hormones and motility among geriatric postoperative patients with gastrointestinal tumors. *J Tradit Chin Med* 2016;36(4):450–5.
- [7] Venara A, Neunlist M, Slim K, Barbieux J, Colas PA, Hamy A, et al. Postoperative ileus: pathophysiology, incidence, and prevention. *J Visc Surg* 2016;153(6):439–46.
- [8] Paulasir S, Kaoutzanis C, Welch KB, Vandewarker JF, Krapohl G, Lampman RM, et al. Nonsteroidal anti-inflammatory drugs: do they increase the risk of anastomotic leaks following colorectal operations?. *Dis Colon Rectum* 2015;58(9):870–7.
- [9] Yang FM, Yao L, Wang SJ, Guo Y, Xu ZF, Zhang CH, et al. Current tracking on effectiveness and mechanisms of acupuncture therapy: a literature review of high-quality studies. *Chin J Integr Med* 2020;26(4):310–20.
- [10] Chi YL, Zhang WL, Yang F, Su F, Zhou YK. Transcutaneous electrical acupoint stimulation for improving postoperative recovery, reducing stress and inflammatory responses in elderly patient undergoing knee surgery. *Am J Chin Med* 2019;47(7):1445–58.
- [11] Huang W, Long W, Xiao J, Zhao G, Yu T. Effect of electrically stimulating acupoint, Zusanli (ST36), on patient's recovery after laparoscopic colorectal cancer resection: a randomized controlled trial. *J Tradit Chin Med* 2019;39(3):433–9.
- [12] Wang H, Yang G, Wang S, Zheng X, Zhang W, Li Y. The most commonly treated acupuncture indications in the United States: a cross-sectional study. *Am J Chin Med* 2018;46(7):1–33.
- [13] Martin CR, Osadchiy V, Kalani A, Mayer EA. The brain-gut-microbiome axis. *Cell Mol Gastroenterol Hepatol* 2018;6(2):133–48.
- [14] Yu YC, Li J, Zhang M, Pan JC, Yu Y, Zhang JB, et al. Resveratrol improves brain-gut axis by regulation of 5-HT-dependent signaling in the rat model of irritable bowel syndrome. *Front Cell Neurosci* 2019;13:30.
- [15] Xue Z, Wu C, Wei J, Xian M, Wang T, Yang B, et al. An orally administered magnololide A ameliorates functional dyspepsia by modulating brain-gut peptides and gut microbiota. *Life Sci* 2019;233:116749.
- [16] Cheng H, Zhang XT, Yan H, Bai LJ, Ai L, Wang FB, et al. Differential temporal neural responses of pain-related regions by acupuncture at acupoint ST36: a magnetoencephalography study. *Chin Med J (Engl)* 2011;124(8):1229–34.
- [17] Bai YF, Gao C, Li WJ, Du Y, An LX. Transcutaneous electrical acupuncture stimulation (TEAS) for gastrointestinal dysfunction in adults undergoing abdominal surgery: study protocol for a prospective randomized controlled trial. *Trials* 2020;21(1):617.
- [18] An LX, Chen X, Ren XJ, Wu HF. Electro-acupuncture decreases postoperative pain and improves recovery in patients undergoing a supratentorial craniotomy. *Am J Chin Med* 2014;42(5):1099–109.
- [19] Yang Y, Zuo HQ, Li Z, Qin YZ, Mo XW, Huang MW, et al. Comparison of efficacy of simo decoction and acupuncture or chewing gum alone on postoperative ileus in colorectal cancer resection: a randomized trial. *Sci Rep* 2017;7:37826.
- [20] Qu LZ, Zhao HC, Shen HX, Yan Y, Ma SJ, Wang KQ. Electroacupuncture of both single- and multi-acupoints promotes recovery of gastrointestinal function in laparoscopic cholecystectomy patients. *Zhen Ci Yan Jiu* 2020; 45(2): 136–40 [Chinese with abstract in English].
- [21] Bragg D, El-Sharkawy AM, Psaltis E, Maxwell-Armstrong CA, Lobo DN. Postoperative ileus: recent developments in pathophysiology and management. *Clin Nutr* 2015;34(3):367–76.
- [22] Hedrick TL, McEvoy MD, Mythen MMG, Bergamaschi R, Gupta R, Holubar SD, et al. American Society for Enhanced Recovery and Perioperative Quality initiative joint consensus statement on postoperative gastrointestinal dysfunction within an enhanced recovery pathway for elective colorectal surgery. *Anesth Analg* 2018;126(6):1896–907.
- [23] Shin KM, Park JE, Lee S, Choi SM, Ahn YC, Lee JW, et al. Effect of Siganu acupuncture on gastrointestinal motility: a randomized, sham-controlled, crossover trial. *Evid Based Complement Alternat Med* 2013;2013:918392.
- [24] Steinhoff MS, von Mentzer B, Geppetti P, Pothoulakis C, Bunnett NW. Tachykinins and their receptors: contributions to physiological control and the mechanisms of disease. *Physiol Rev* 2014;94(1):265–301.
- [25] Chen J, Tu Q, Miao S, Zhou Z, Hu S. Transcutaneous electrical acupoint stimulation for preventing postoperative nausea and vomiting after general anesthesia: a meta-analysis of randomized controlled trials. *Int J Surg* 2020;73:57–64.
- [26] Fu C, Wu T, Shu Q, Song A, Jiao Y. Acupuncture therapy on postoperative nausea and vomiting in abdominal operation: a Bayesian network meta-analysis. *Medicine (Baltimore)* 2020;99(23):e20301.